# Connect to Oracle Database from VS Code

*Participant quick-start guide • Oracle SQL Developer for VS Code*

Scope: This guide covers only creating, testing, and using the Oracle database connection inside Visual Studio Code. It assumes the Oracle Cloud database and wallet have already been provisioned and the required VS Code extension is installed.

## 1. Before you start

- Visual Studio Code is installed and opens successfully.
- The Oracle SQL Developer extension for VS Code is installed and enabled.
- The database wallet ZIP provided by the instructor has been extracted to a folder on your computer.
- You have been given your own database username and password, plus the approved TNS service name (for this workshop: sqlworkshop_medium).

Wallet reminder: select the extracted wallet folder when prompted by the extension. The folder should contain files such as tnsnames.ora, sqlnet.ora, cwallet.sso and/or ewallet.p12. Keep the wallet files together; do not rename or edit them.

## 2. Open the Oracle connection panel

1. Launch VS Code.
2. Select the Oracle / SQL Developer icon in the Activity Bar. If it is not visible, open the Command Palette (Ctrl+Shift+P) and search for Oracle or SQL Developer commands.
3. Open the Connections view and choose the option to create a new connection (often shown as a + icon or New Connection).

## 3. Enter the connection details

Use the fields shown by the installed extension. Labels can vary slightly by extension version.

| Field | What to enter |
|---|---|
| **Connection name** | A friendly label, e.g. Workshop Medium |
| **Username** | Your assigned database username (for example, SQLSTUDENT01) |
| **Password** | The database password supplied to you |
| **Connection type / configuration** | Choose Cloud Wallet / Wallet, if the extension presents this option |
| **Wallet location** | Browse to the extracted wallet folder on your computer |
| **TNS / Service name** | sqlworkshop_medium |

Do not use the wallet ZIP password as the database password. They are different credentials. Use only the database account assigned to you; do not share another participant's credentials.

## 4. Test and save the connection

4. Select Test (or Test Connection).
5. Wait for a success message before proceeding. The first connection may take a short while.
6. If the test succeeds, choose Connect / Save, depending on the buttons displayed.

7. Confirm that the saved connection appears in the Connections list. Expand it to browse database objects if permitted.

## 5. Run a test query in VS Code

Open a new SQL file (File → New Text File, then save it with a .sql extension), or open an existing .sql file. Enter:

```
SELECT USER FROM dual;

SELECT banner FROM v$version;
```

Select the saved connection as the active connection for the SQL editor, if required by the extension. Run the statement using the extension's Run Statement command or its displayed shortcut/button. The first query should return the database username you connected as. The version query may require permission; if it is denied, use SELECT USER FROM dual; as the basic connectivity check.

## 6. Troubleshooting

| Symptom | What to check |
|---|---|
| **Connection test fails immediately** | Recheck the username, database password, and selected TNS service name. |
| **TNS alias / service not found** | Confirm that the wallet folder is extracted and the extension is pointed to the correct folder; verify that sqlworkshop_medium is listed in tnsnames.ora. |
| **Wallet / TLS / certificate error** | Keep all wallet files together and select the folder containing them. If the wallet is incomplete or corrupted, request a fresh wallet from the instructor. |
| **Authentication failed** | The database password may be incorrect or expired, or the account may be locked. Confirm the assigned credentials with the instructor. |
| **Connection takes a while** | Wait briefly, then test again. Check that the computer has internet access and that no corporate network or VPN rule is blocking the connection. |
| **Query does not run against the connection** | Ensure the SQL editor is associated with the saved connection, then run the statement again. |

## 7. Security checklist

- Use your assigned database account; do not use the instructor's ADMIN account unless explicitly authorized.
- Never put passwords in a shared .sql file, screenshots, or source control.
- Do not upload wallet files to a public repository or share them outside the approved workshop channel.
- When finished on a shared computer, disconnect the session and follow the instructor's instructions for removing saved credentials.

**Connection checklist:** Extension installed → wallet folder selected → sqlworkshop_medium selected → own credentials entered → Test succeeds → SELECT USER FROM dual; returns your username.